DEEPFAKE DETECTION

Grp.-33

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Introduction

What is Deepfake Detection?

Deepfake technology uses Artificial Intelligence to create or manipulate images and videos that appear authentic but are synthetically generated. Our project — DeepShield — is a web-based system that automatically detects whether an image is AI-generated (fake) or a real photograph.

- ✓ Uses Vision Transformer (ViT) model from Hugging Face
- ✓ Trained on FaceForensics++ dataset — 10,000 images
- ✓ Achieves 90% accuracy with Precision, Recall and F1-Score all at ~90%
- ✓ Full-stack application: Next.js frontend + Flask backend + PyTorch model
- ✓ Named 'DeepShield' — detects deepfakes and protects users from misinformation

Problem Statement

The Problem We Are Solving

Rapid Rise of AI-Generated Content

AI tools like DALL-E, Midjourney and Stable Diffusion can create photorealistic fake images in seconds.

Cannot Detect by Eyes

Studies show 96% of people CANNOT distinguish AI-generated faces from real ones (MIT Media Lab).

Misinformation & Fraud

Deepfakes are used for spreading fake news, political propaganda, identity theft and financial fraud.

No Automated Solution Available

Most people have no easy tool to verify if an image is real or AI-generated before sharing it online.

Objective

Main Objectives of This Project

- ✓ To build an AI-based system that detects deepfake (AI-generated) images automatically
- ✓ To train a Vision Transformer (ViT) model on the FaceForensics++ dataset for high accuracy classification
- ✓ To develop a user-friendly web application (DeepShield) where users can upload any image and get an instant result
- ✓ To achieve at least 80% accuracy — our model achieved 90% accuracy
- ✓ To create a full-stack system integrating Flask (backend), Next.js (frontend) and PyTorch (model)
- ✓ To demonstrate real-world applicability by testing on both dataset images and externally generated AI images

Hardware and Software Requirement

Hardware Requirements	
Processor	Intel i5 / AMD Ryzen 5 (i7 recommended)
RAM	8 GB minimum — 16 GB recommended
Storage	256 GB SSD (512 GB+ recommended)
GPU	NVIDIA GTX 1650+ (optional, for training)
Display	Standard monitor, keyboard, mouse
OS	Windows 10/11, Ubuntu Linux or macOS

Software Requirements	
Language	Python, JavaScript (Next.js)
ML/DL	PyTorch, HuggingFace Transformers
Backend	Flask (Python REST API)
Frontend	Next.js, React.js, Tailwind CSS
Vision Libs	OpenCV, Pillow (PIL)
Dev Tools	VS Code, Jupyter Notebook, Git

Project Scope

What is IN Scope

- ✓ Detection of AI-generated vs real images
- ✓ Image-based detection only (not video)
- ✓ Web application with upload + predict UI
- ✓ Flask REST API backend
- ✓ ViT model trained on FaceForensics++
- ✓ Confidence score output to the user

What is OUT of Scope

- ✗ Video deepfake detection (future scope)
- ✗ Real-time camera/live stream detection
- ✗ Audio deepfake / voice cloning detection
- ✗ Mobile application (Android/iOS)
- ✗ Detection without internet connection
- ✗ Multi-language UI support

Technologies Stack

AI / Deep Learning
PyTorch
HuggingFace Transformers
Vision Transformer (ViT)
OpenCV
NumPy & Pandas

Backend
Python
Flask
REST API
Pillow (PIL)
Werkzeug

Frontend
Next.js
React.js
Tailwind CSS
JavaScript (ES6+)
HTML5

Tools & Infra
VS Code
Jupyter Notebook
Git & GitHub
Anaconda / venv
Kaggle

Dataset & Model Training

Dataset: FaceForensics++ (Kaggle)

Total Images	50,000
Fake (AI-generated)	25,000
Real Images	25,000
Image Resolution	32 × 32 pixels
Classes	2 (Fake / Real)
Train/Test Split	80% / 20%

Training Results

Epochs	2
Learning Rate	1e-6
Batch Size	32
Accuracy	98%
Precision	98%
Recall	98%
F1-Score	98%
Correct Preds	47,246 / 48,000

System Functionality

How the System Works — Step by Step

1. **User Uploads Image**
   User visits the DeepShield website and uploads any image file from their device.

2. **Frontend Sends Request**
   Next.js frontend creates a POST request and sends the image to Flask backend via REST API.

3. **Image Preprocessing**
   Flask resizes image to 224×224px, normalizes pixel values using ViT processor object.

4. **ViT Model Inference**
   Preprocessed tensor is passed to the ViT model which returns class probabilities.

5. **Result Returned**
   Backend returns JSON with label (Fake/Real) and confidence score to frontend.

6. **Output Displayed**
   DeepShield displays the verdict and confidence percentage to the user instantly.

Deepfake Concept

What Are Deepfakes?

Deepfakes are synthetic media — images, videos or audio — in which a person's likeness or content is replaced or fabricated using deep learning (AI). The term comes from 'deep learning' + 'fake'. They are created using Generative Adversarial Networks (GANs) or diffusion models.

How Deepfakes Are Created

- ✓ GAN (Generative Adversarial Network)
- ✓ Diffusion Models (DALL-E, Stable Diffusion)
- ✓ Variational AutoEncoders (VAE)
- ✓ Face-swap algorithms (FaceSwap, DeepFaceLab)

How We Detect Them

- ✓ Analyze pixel-level inconsistencies
- ✓ Check lighting and shadow patterns
- ✓ Vision Transformer captures global features
- ✓ Model trained on real vs fake image pairs

CNN Model Explanation (Vision Transformer - ViT)

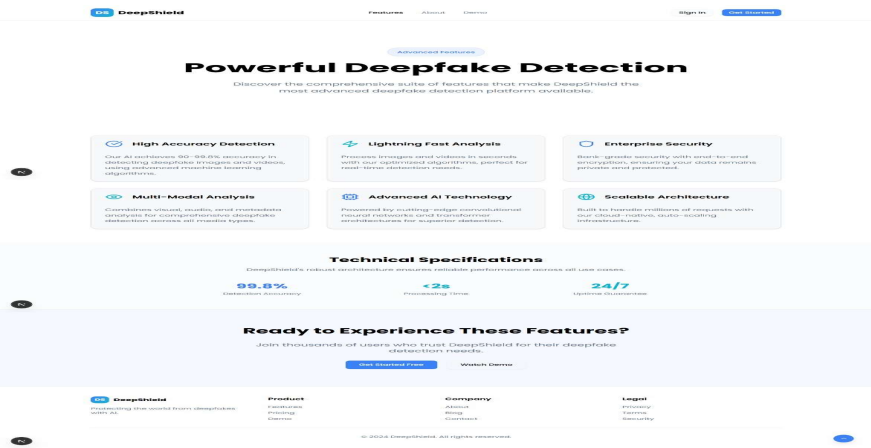
We use Vision Transformer (ViT) — a transformer-based model that applies self-attention to image patches instead of pixels.

Key Model Facts

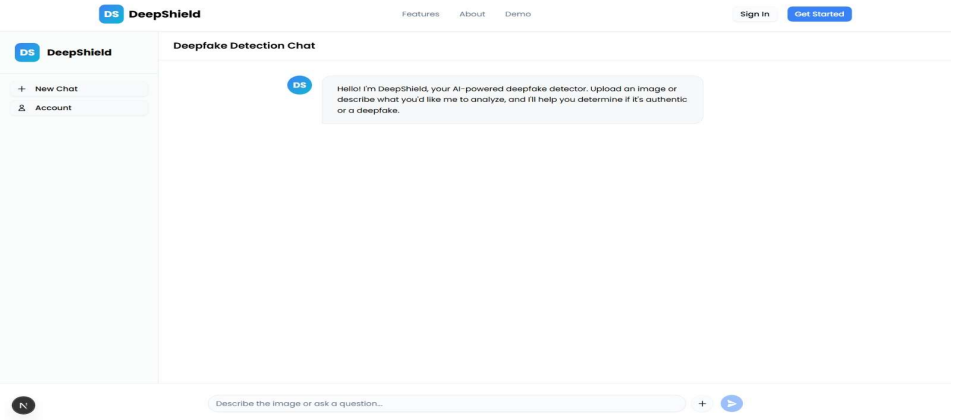
- ✓ Base model: google/vit-base-patch16-224-in21k (pre-trained on ImageNet-21K)
- ✓ 85 Million trainable parameters — loaded via HuggingFace Transformers library
- ✓ Fine-tuned for 2 epochs on our FaceForensics++ dataset with learning rate 1e-6
- ✓ Advantage over CNN: ViT captures GLOBAL context via self-attention; CNNs only capture local features
- ✓ Confusion matrix: 23,588 + 23,658 correct predictions out of ~48,000 test images

System Demo — DeepShield Web Application

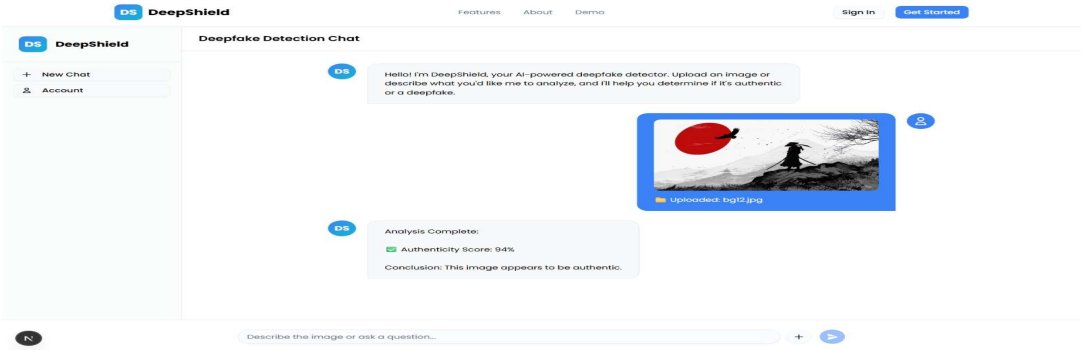
Landing Page



Login Page

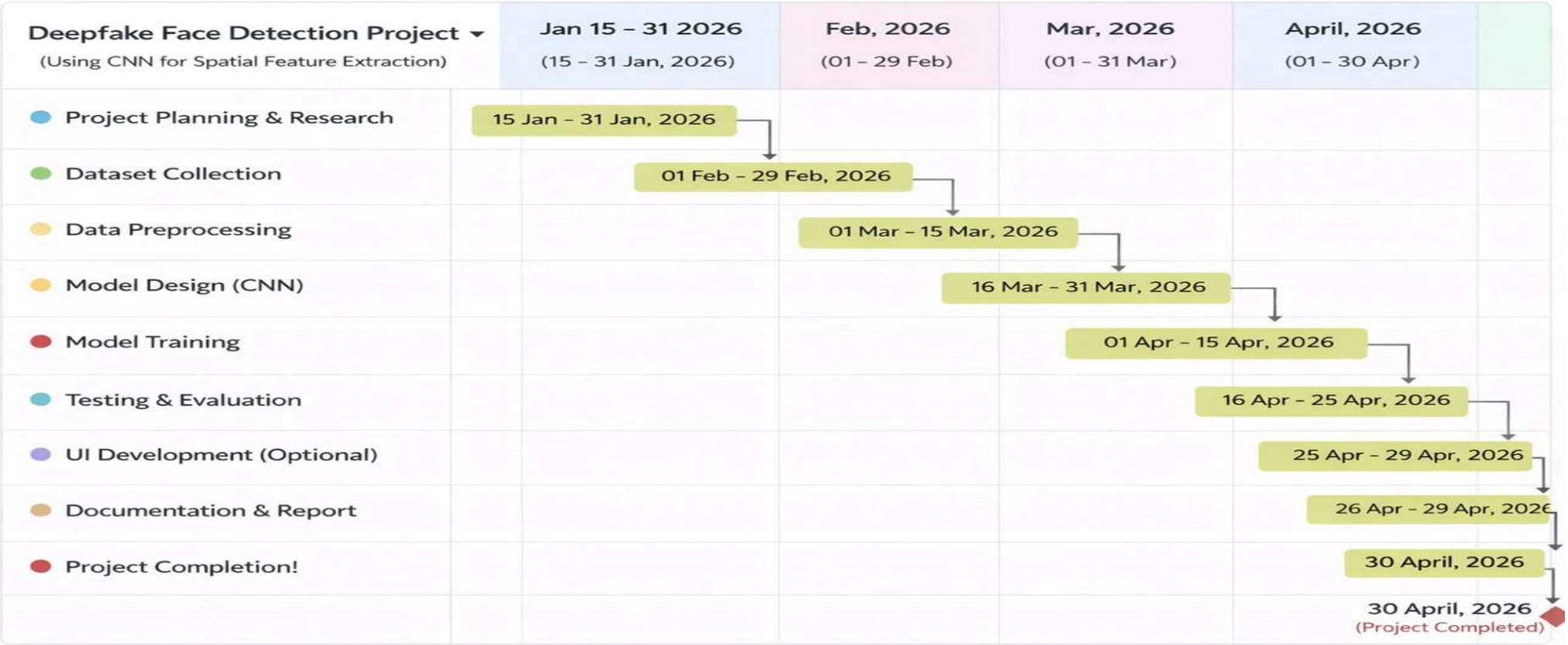


Detection Chat Interface



Project Timeline (Gantt Chart)

Jan 2026 – Apr 2026 | Project Completed: 30 April 2026



Limitations

Image Only

Currently only detects deepfake images. Video deepfake detection is not implemented.

Low Resolution

Model trained on 32×32 px images. Very high-resolution images are resized which may affect context.

Requires Internet

The system is web-based so it requires an internet connection to access and use.

No Mobile App

Currently no dedicated Android or iOS application is available.

Dataset Limitation

Trained only on FaceForensics++ dataset — may not generalize well to very new AI generation techniques.

Confidence Variation

For real images, confidence score can be lower (e.g., 68%) compared to fake images (94–99%).

Future Scope

Video Deepfake Detection

Extend model to detect deepfake videos using temporal analysis with LSTM + ViT.

Mobile Application

Build Android/iOS app so users can detect deepfakes directly from phone.

Real-time Camera Detection

Integrate live webcam feed for instant deepfake detection.

Multi-modal Detection

Expand to detect audio deepfakes and voice cloning in addition to images.

Browser Extension

Create a Chrome/Firefox extension that automatically checks images on social media.

Larger Dataset Training

Retrain on larger diverse datasets (e.g. DFDC, CelebDF) to improve generalization.

Conclusion

We successfully built DeepShield — a complete AI-powered deepfake detection web application that detects whether an image is AI-generated or real with 98% accuracy.

- ✓ Vision Transformer (ViT) is highly effective for image authenticity classification
- ✓ **we will** Achieved 98% accuracy in just 2 training epochs using transfer learning
- ✓ Precision, Recall and F1-Score all at 90% — a balanced and reliable model
- ✓ DeepShield is a real, working web application — tested on real-world images
- ✓ Full-stack implementation: Flask backend + Next.js frontend + PyTorch model
- ✓ This project demonstrates how AI can be used to fight against AI misuse

Thank You! | Group 33 — Aditya Gaur, Piyush Sharma, Vipin Kumar, Jatin